

Catalytic competence bounds the decay rates of a thermally formed radical pair

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(Dated: 6 October 2026)

A flavoenzyme could respond to the geomagnetic field through the radical-pair mechanism only if its catalytic cycle passes through a radical pair that lives long enough for a 50 μT field to act. For a pair formed thermally from a closed-shell enzyme–substrate complex, returning to it only through the singlet and carrying the catalytic flux, we show that detailed balance bounds both decay rates from below: the singlet back-transfer rate must exceed $4k_{\text{cat}}e^{\Delta/k_B T}$ and the spin-independent onward rate $k_{\text{cat}}e^{\Delta/k_B T}$, where Δ is the free energy of the pair. The second bound rests on an inequality, $\Phi_P \leq 4k_P/k_S$, that holds for any spin Hamiltonian and any trace-preserving, unital, positivity-preserving relaxation and cannot be improved over that class; neither bound involves electronic couplings or reorganization energies. In a numerical search over a flavin–aminium spin model with density-functional hyperfine and dipolar couplings, the largest field effect found among pairs that satisfy both bounds for $k_{\text{cat}} = 1 \text{ s}^{-1}$ falls below 0.1 % at $\Delta = 0.47 \text{ eV}$ (0.50 eV with anisotropic hyperfine tensors) and stays below up to 0.61 eV, the grid limit; electron spin relaxation does not raise these thresholds. Measured flavin potentials of the monoamine oxidases, with a dialkylamine potential as a substrate proxy, place an amine-to-flavin pair 0.86–1.44 eV above the closed-shell state in water; continuum estimates for the active site bring it within the range of a measurable effect only in the most favourable case, about 0.49 eV.

I. INTRODUCTION

The radical-pair mechanism (RPM) is the best-established way in which a magnetic field as weak as the geomagnetic field ($\sim 50 \mu\text{T}$) can alter the yield of a chemical reaction.^{1,2} It underlies the leading model of avian magnetoreception, in which photoexcitation of the flavin in cryptochrome creates a spin-correlated flavin–tryptophan radical pair.^{3–5} Whether the same mechanism could act in flavoenzymes that work in the dark, such as the monoamine oxidases that degrade neurotransmitters, has been raised repeatedly.^{6,7}

General limits on weak-field effects are known. Electron spin relaxation is expected to limit a primary RPM effect in biochemistry to 10^{-3} – 10^{-4} or less,⁸ and exchange and dipolar couplings between the radicals suppress singlet–triplet interconversion unless the radicals are far apart.⁹ Asymmetric reaction schemes, in which only the singlet pair recombines while both spin states react onward, relax the second limit and allow closely bound pairs to respond.¹⁰ For a pair that recombines by non-adiabatic electron transfer, the ratio of the reactive exchange coupling to the recombination rate is fixed by the activation energy of the transfer.¹¹ These results constrain the spin dynamics of a pair once its rates are given. They do not say which rates a pair can have when an enzyme forms it *thermally* and turns over through it.

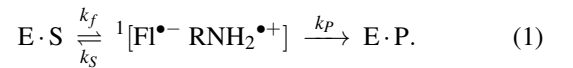
Here we derive that constraint. For a pair formed from a closed-shell complex that must carry the catalytic flux, detailed balance bounds both of its decay rates from below in terms of the driving force Δ and the turnover number. Without spin dynamics both bounds follow from the steady-state turnover through a pre-equilibrium intermediate (Sec. III). Singlet–triplet mixing can raise the onward yield above its spin-free value, because triplet pairs cannot return, and so weakens the bound on the onward rate. We show that it weakens it by at most a factor of four, for any spin Hamiltonian and

any admissible relaxation, and that no smaller factor holds for that whole class. We then combine it with a numerical search for the largest field effect at given rates in a flavin–aminium spin model, state how that search depends on the model, and estimate where the monoamine oxidase pair lies from the measured flavin potentials of the enzyme and a proxy for the amine potential.

II. REACTION SCHEME AND SPIN MODEL

A. Scheme and observable

We consider the single-electron-transfer (SET) scheme proposed for amine oxidation by flavoenzymes:¹²



Electron transfer from the closed-shell complex forms the pair in its singlet state; only the singlet pair returns by back electron transfer (rate k_S), and singlet and triplet pairs react onward with a spin-independent rate k_P , for example by α -C–H deprotonation of the aminium radical. Aminium radical cations can also deprotonate at nitrogen: dialkylaminium radicals in water have $pK_a = 5.3$ – 6.8 , and the diethylaminium radical deprotonates predominantly at nitrogen under the conditions studied.¹³ What follows requires only that the onward step, whatever its chemistry, is spin independent. The yield of onward reaction per pair formed is $\Phi_P = k_P \int_0^\infty \text{Tr} \rho dt$, and the magnetic field effect is $\text{MFE}_P = [\Phi_P(B) - \Phi_P(0)]/\Phi_P(0)$ at $B = 50 \mu\text{T}$. The spin density operator of a pair born in the singlet evolves under a master equation with Haberkorn reaction terms,^{11,14}

$$\begin{aligned} \dot{\rho} = \mathcal{L}\rho &= -2\pi i[H, \rho] - \frac{k_S}{2}\{P_S, \rho\} - k_P\rho + \mathcal{D}\rho, \\ \rho(0) &= P_S/N, \end{aligned} \quad (2)$$

with P_S the singlet projector, $N = \text{Tr}P_S$ the nuclear-spin dimension and $\mathcal{D}$ a spin-relaxation superoperator. We assume that $\mathcal{D}$ is trace preserving, $\text{Tr}\mathcal{D}(X) = 0$, and generates a positivity-preserving evolution, as a trace-preserving Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) generator^{15,16} does, and that it is unital, $\mathcal{D}(I) = 0$, as single-electron dephasing, the reactive singlet–triplet dephasing of ref 11 and any relaxation that drives the spins towards equal populations at high temperature are. The reaction terms also assume that the nuclear motion driving electron transfer is fast compared with the spin dynamics, the incoherent-recombination approximation of ref 11. Yields are obtained exactly: without relaxation from the Sylvester equation for $\int \rho dt$, solved by the Bartels–Stewart method,¹⁷ and with relaxation by the same Sylvester solve inside a GMRES iteration¹⁸ (Supporting Information).

B. Spin Hamiltonian and conventions

In frequency units,

$$H = \omega_B \hat{n}_B \cdot (\mathbf{S}_1 + \mathbf{S}_2) + \sum_k \mathbf{S}_{e(k)} \cdot \mathbf{A}_k \cdot \mathbf{I}_k + J \mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_1 \cdot \mathbf{T} \cdot \mathbf{S}_2, \quad (3)$$

with $\omega_B = g\mu_B B/h$ and hyperfine tensors $\mathbf{A}_k$. $J = E_T - E_S$ is the full singlet–triplet gap; in the convention of Efimova and Hore⁹ $J = -2J_{\text{EH}}$. $\mathbf{T}$ is the traceless dipolar tensor; for an axial tensor $\mathbf{S}_1 \cdot \mathbf{T} \cdot \mathbf{S}_2 = 2D[(\mathbf{S}_1 \cdot \hat{n})(\mathbf{S}_2 \cdot \hat{n}) - \mathbf{S}_1 \cdot \mathbf{S}_2/3]$, so that with the field along $\hat{n}$ the $T_{\pm}$ and T_0 levels are split by D ($D = -77.9$ MHz for point dipoles 1 nm apart⁹). Both conventions are verified numerically in the code.

C. Magnetic parameters

The reference model has four nuclei: flavin N5 (523 μT) and N10 (189 μT), each as an isotropic spin-1 nucleus,¹⁹ although both couplings are strongly anisotropic; anisotropic tensors are examined in Sec. VI. The model also has the two largest couplings of the benzylaminium radical cation, 257 and 196 MHz ($\alpha\text{-CH}_2$ protons; B3LYP^{20,21}/def2-TZVP²² in PySCF²³ on GFN2-xTB geometries²⁴). The dipolar tensor of the contact pair is the point-dipole tensor summed over the Löwdin²⁵ spin populations of the two radicals, as Efimova and Hore did for cryptochrome,⁹ for methylamine stacked on the flavin si face at an N5–N distance of 3.5 Å, with $D = -843$ MHz and $E = -26$ MHz. Because the pair geometry is not known, the search below scales this tensor by any factor $s \in [0, 3]$, from no dipolar coupling to three times the contact value. Richer nuclear models and spin relaxation are examined in Sec. VI.

III. CATALYTIC COMPETENCE BOUNDS BOTH DECAY RATES

Let Δ be the free energy of the radical pair relative to E·S, summed over its four spin states, so that, with spin energies

negligible against $k_B T$, the singlet pair lies $\Delta_S = \Delta + k_B T \ln 4$ above E·S ($k_B T \ln 4 = 0.037$ eV at 310 K) and detailed balance for the singlet channel gives $k_f = k_S e^{-\Delta_S/k_B T}$. The turnover carried by the pair is $k_f \Phi_P$ per E·S complex, and two inequalities bound Φ_P . Trivially $\Phi_P \leq 1$. Less obviously,

$$\Phi_P \leq 4k_P/k_S \quad (4)$$

for any spin Hamiltonian and any $\mathcal{D}$ with the properties above (trace preserving, unital, positivity preserving). Indeed, for $k_P > 0$ the resolvent $\mathcal{R} = (-\mathcal{L})^{-1} = \int_0^\infty e^{\mathcal{L}t} dt$ exists and maps positive operators to positive operators, and unitality gives $-\mathcal{L}(I) = k_S P_S + k_P I$; applying $\mathcal{R}$, $k_S \mathcal{R}(P_S) = I - k_P \mathcal{R}(I) \preceq I$, and taking the trace, $\Phi_P = (k_P/N) \text{Tr} \mathcal{R}(P_S) \leq 4k_P/k_S$. Without spin dynamics $\Phi_P = k_P/(k_S + k_P)$, a quarter of the bound when $k_P \ll k_S$; the factor of four is approached when singlet–triplet mixing is complete, as under fast depolarizing relaxation, where $\Phi_P k_S/4k_P = 0.999996$ even without hyperfine couplings (Supporting Information). It therefore cannot be reduced for the class as a whole, although a given system may stay well below it. Over 300 random spin systems with exchange, dipolar and hyperfine couplings and, in part, electron spin relaxation, the largest value of $\Phi_P k_S/4k_P$ is 0.9986 (Supporting Information).

Requiring the turnover $k_f \Phi_P$ to be at least k_{cat} gives

$$k_S \geq 4k_{\min}(\Delta) \quad \text{and} \quad k_P \geq k_{\min}(\Delta), \quad (5)$$

$$k_{\min}(\Delta) \equiv k_{\text{cat}} e^{\Delta/k_B T},$$

the first from $\Phi_P \leq 1$ and the second from eq 4. Eq 5 is a statement about the generator of eq 2; whether a strongly coupled contact pair is described by such a generator is a separate question, taken up in Sec. VII. Without spin dynamics the turnover is exactly $e^{-\Delta_S/k_B T}/(1/k_S + 1/k_P)$, which requires $k_P \geq 4k_{\min}$ as well; the spin dynamics relaxes only this second condition. Eq 5 contains no electronic coupling, no reorganization energy and no assumption about the chemistry of the onward step. If, in addition, we assume that no elementary rate exceeds $\nu_n \approx 10^{13}\text{--}10^{14} \text{ s}^{-1}$, at or above the transition-state prefactor $k_B T/h = 6.5 \times 10^{12} \text{ s}^{-1}$ at 310 K,²⁶ the bound on k_S requires a pair that carries $k_{\text{cat}} = 1\text{--}10^3 \text{ s}^{-1}$ to have $\Delta \leq k_B T \ln(\nu_n/4k_{\text{cat}}) = 0.58\text{--}0.82$ eV.

A radical pair on a dead-end branch, which does not lead to product, is a separate case. It carries no flux, so eq 5 does not apply to it, but it holds enzyme, and a field that changes its population changes the free enzyme. With $k_P = 0$ the resolvent need not exist (triplet states that never reach the singlet are stationary), but the identity $k_S \int_0^t e^{\mathcal{L}_0 u} (P_S) du = I - e^{\mathcal{L}_0 t} (I) \preceq I$, with $\mathcal{L}_0$ the generator at $k_P = 0$, bounds the integrated singlet-born population by $4N/k_S$ for all t . The steady-state population of the branch is therefore at most $4k_f/k_S = e^{-\Delta/k_B T}$ per E·S complex, and so is the fraction of enzyme through which a field can act on it.

IV. THE LARGEST FIELD EFFECT FOUND AT GIVEN RATES

The exchange coupling of a contact pair is the sum of a reactive contribution from the electron transfer that sets k_S ¹¹

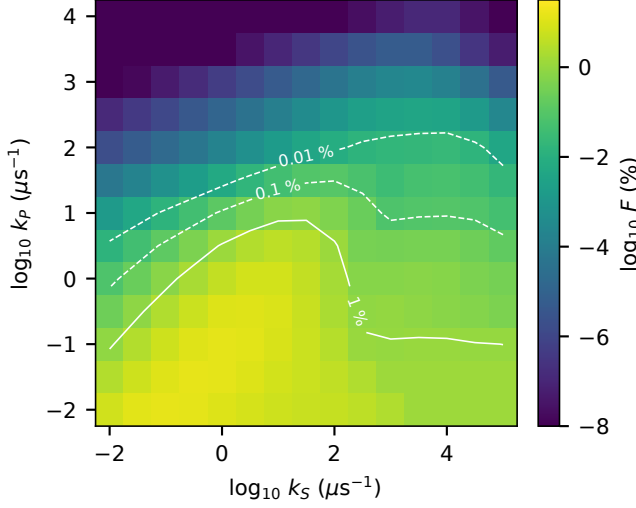


FIG. 1. Largest magnetic field effect on the onward yield found by the search over the exchange coupling, the field direction and the dipolar scale $s \in [0, 3]$, as a function of the singlet back-transfer rate k_S and the spin-independent onward rate k_P . Reference four-nucleus model, coherent spin dynamics, $B = 50 \mu\text{T}$.

and a direct contribution that the rates do not fix. We therefore treat J as free and search, for given (k_S, k_P) , for

$$F^*(k_S, k_P) = \sup_{J, \hat{n}_B, s \in [0, 3]} |\text{MFE}_P|. \quad (6)$$

We write $F \leq F^*$ for the largest value the search finds. The search scans J on a grid that is dense near zero and extends to $\max(4000 \text{ MHz}, 5(k_S + k_P)/2\pi)$, for $s = 0, 0.3, 1$ and 3 and 25 field directions over a full hemisphere, refines each of the three largest local maxima in J , and then polishes the four best candidates by a simplex search in $(J, \hat{n}_B, s)$. Because the response varies sharply with s at small s (Supporting Information), a second pass scans s on a fine logarithmic grid (41 values) at a coarse J grid and polishes again. F is the larger of the two passes. The second pass raised 151 of the 195 cells, by up to a factor of 3.6, but changed no row maximum by more than 0.0 %; tripling the density of the s grid at the cells that set the thresholds, and at the cell the second pass raised most, changed F by a factor of at most 1.018. No maximum lies at the edge of the J range (0 of 195 cells).

Figure 1 shows F over $k_S = 10^{-2}$ – $10^5 \mu\text{s}^{-1}$ and $k_P = 10^{-2}$ – $10^4 \mu\text{s}^{-1}$. The onward rate controls the response: $\max_{k_S} F$ is 14 % at $k_P = 10^{-2} \mu\text{s}^{-1}$, 0.023 % at $10^2 \mu\text{s}^{-1}$ and $1.2 \times 10^{-7} \%$ at $10^4 \mu\text{s}^{-1}$. A fast singlet back-transfer does not by itself remove the effect, because the triplet pair cannot return and is drained only by k_P ,¹⁰ but a fast spin-independent decay removes it for every k_S, J and s searched.

V. FIELD EFFECTS OF A COMPETENT PAIR

Combining eq 5 with the field effect defines

$$G^*(\Delta) = \sup \{ F^*(k_S, k_P) : k_S \geq 4k_{\min}(\Delta), k_P \geq k_{\min}(\Delta) \}. \quad (7)$$

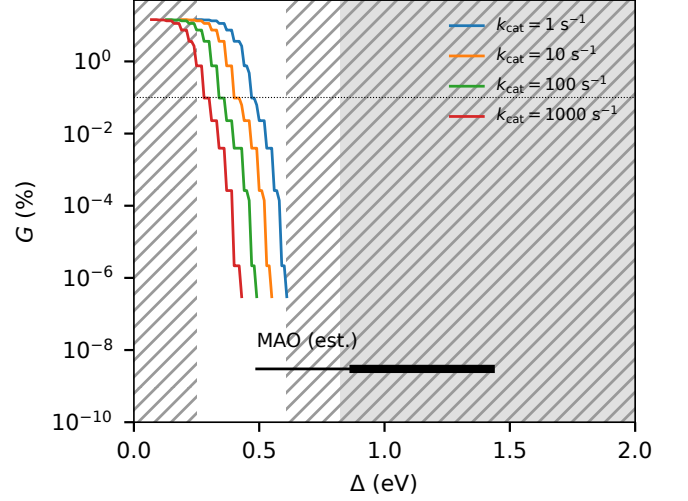


FIG. 2. Largest magnetic field effect found by the grid procedure among pairs that can carry the turnover, $G(\Delta)$, for $k_{\text{cat}} = 1, 10, 100$ and 10^3 s^{-1} in the reference model. Hatched: G not defined for $k_{\text{cat}} = 1 \text{ s}^{-1}$ because the required rates lie outside the computed grid. Shaded: no pair carries $k_{\text{cat}} = 1 \text{ s}^{-1}$ if no rate exceeds $\nu_n = 10^{14} \text{ s}^{-1}$. Bar: driving force of an amine-to-flavin pair in monoamine oxidase estimated from measured flavin potentials and a proxy amine potential, in water (thick) and extended to the most favourable continuum correction for the active site (thin) (Sec. VII).

The region in braces is defined by necessary conditions: it contains every pair that can carry the turnover, but not every pair in it can (whether it does also depends on Φ_P and on the occupancy of E·S). G^* is therefore an upper bound on the field effect of a competent pair, taken over a larger set. We evaluate it by a procedure on the rate grid and call the result G . The required rates are rounded down to the nearest grid rates, so that rates just above them are represented by the cell below; on the threshold row ($\Delta = 0.46 \text{ eV}$, $k_P = 31.6 \mu\text{s}^{-1}$) the off-grid rates $k_S = 126$ – $379 \mu\text{s}^{-1}$ give at most 0.0924 % against $G = 0.0965 \%$ (Supporting Information). G is still a finite-search value, not a proven bound. Because the permitted region only shrinks as Δ grows, G is finally replaced by its running maximum from larger Δ . Where the required rates fall outside the computed grid, G is not reported: below it ($\Delta < 0.25 \text{ eV}$ for $k_{\text{cat}} = 1 \text{ s}^{-1}$) the permitted region includes rates slower than any computed, and above it ($\Delta > 0.61 \text{ eV}$) the required rates exceed the computed ones.

Figure 2 shows G for $k_{\text{cat}} = 1$ – 10^3 s^{-1} . Because k_P is sampled every half decade, the thresholds below are resolved to about 0.03 eV. For $k_{\text{cat}} = 1 \text{ s}^{-1}$, G falls below 0.1 % at $\Delta = 0.47 \text{ eV}$ and below 0.01 % at $\Delta = 0.53 \text{ eV}$, and stays below up to 0.61 eV, beyond which it is not computed; for $k_{\text{cat}} = 10^3 \text{ s}^{-1}$ the 0.1 % threshold is 0.28 eV. A pair near thermoneutrality is not excluded: at small Δ competence permits slow decay and the field effect can be several percent.

VI. DEPENDENCE ON THE SPIN MODEL

F and G belong to the reference model, and they change with it. In the reference model N5 and N10 are isotropic. We repeated the full search, both passes and the edge rule, with anisotropic tensors for both nuclei (density-functional spin-dipolar parts in the frame of the dipolar tensor) on every cell that the thresholds use on the rows $k_P = 10\text{--}10^{2.5} \mu\text{s}^{-1}$ with $k_S \geq k_P$ (30 cells; Supporting Information). Individual cells changed by factors of up to 7.4, and the row maxima by factors of 0.76–3.11. With each of these cells replaced by the larger of its two values, the 0.1 % threshold for $k_{\text{cat}} = 1 \text{ s}^{-1}$ is 0.50 eV, and 0.31 eV for $k_{\text{cat}} = 10^3 \text{ s}^{-1}$. Rows with $k_P \geq 10^3 \mu\text{s}^{-1}$ keep their reference values, which are below $2.6 \times 10^{-4} \%$. Adding the aminium ^{14}N or the two largest flavin methyl protons was examined only by searching again around the optimum of the reference row-maximum cell (Supporting Information); such a local search can miss larger values elsewhere in the row, and we do not use it to move the thresholds.

Electron spin relaxation was computed directly in the four-nucleus model. Every cell that the thresholds use on these rows ($k_S \geq k_P$, 90 searches) was searched again with single-electron dephasing, $T_2^e = 10 \text{ ns}$, 100 ns and $1 \mu\text{s}$, using a solver that does not build the Liouvillian (Supporting Information). Relaxation changed individual cells by factors of 0.11–4.94, so it does not reduce F everywhere, but it lowered every row maximum: the largest value found with relaxation in each row is 0.84–0.99 times the coherent one. With each cell replaced by the larger of its coherent and relaxed values, the 0.1 % threshold for $k_{\text{cat}} = 1 \text{ s}^{-1}$ is 0.47 eV and the 0.01 % threshold 0.53 eV.

VII. WHERE MONOAMINE OXIDASE LIES

The potentials of the covalent flavin in the monoamine oxidases have been measured. Chemical reduction proceeds in two steps, the first giving the semiquinone and each requiring two electron equivalents of the enzyme; the oxidized/semiquinone midpoint potential is -159 mV in MAO A and -167 mV in MAO B,²⁷ and the flavin radical of MAO A is anionic,²⁸ the radical of the scheme above. We take these midpoints as the one-electron flavin couple. For the amine, dialkylamine radical cations in water have $E^\circ(\text{R}_2\text{NH}^{\bullet+}/\text{R}_2\text{NH}) = 1.30 \pm 0.05 \text{ V}$ vs NHE, a value its authors give as a general approximation;¹³ we are not aware of a measured potential for a primary alkylamine or benzylamine radical cation in water and use this value as a proxy. Its uncertainty is probably larger than stated. The dialkylamine values come close to the practical limit of cyclic voltammetry in water, and follow-up reactions of the radical cations can shift peak potentials negative by up to about 0.2 V, so that the standard potentials may lie higher; a computed value of 0.80 V for dimethylamine, much lower than the measured ones, has also been reported.¹³ The direction of the error for primary amines is not established either: the solvation difference between amine and radical cation grows with the gas-phase ionization energy,¹³ which would partly offset the higher ioniza-

tion energies of primary amines. With the Coulomb attraction of the contact pair in water (0.03–0.05 eV for 6–3.5 Å) this places the pair at $\Delta \approx 1.41\text{--}1.44 \text{ eV}$ in water.

These numbers describe the pair in water. In the enzyme, Δ follows from a thermodynamic cycle that adds, to the potentials in water, the free energies of transferring the neutral and charged species from water into the active site and the Coulomb attraction of the pair there. Neither transfer free energy is known. Two continuum limits bracket their effect (Supporting Information). Moving the aminium alone into a medium of effective dielectric constant 4–10, with a Born radius of 2.81 Å calibrated to the measured solvation-energy difference of dimethylamine and its radical cation,¹³ while leaving the flavin anion solvated, changes Δ by -0.37 to $+0.04 \text{ eV}$; moving both ions with the same density-functional continuum protocol changes it by $+0.01$ to $+0.56 \text{ eV}$.

Substrate binding changes the flavin potential: in the presence of substrates the midpoint potential for full reduction shifts by up to $+0.5 \text{ V}$, and no semiquinone forms.²⁷ If the absence of semiquinone reflects equilibrium, the oxidized/semiquinone potential lies at or below the full-reduction midpoint, which for MAO A (-159 and -262 mV without substrate) gives at most $+0.29 \text{ V}$, a shift of at most 0.45 V. We shift the one-electron couple by the full $+0.5 \text{ V}$ as an upper bound on this reading. The pair then lies at $\Delta \approx 0.86\text{--}0.99 \text{ eV}$ in water, including the uncertainty of the amine proxy. This is above the competence limit $\Delta \leq 0.82 \text{ eV}$ of Sec. III, so in water such a pair could not carry the turnover. With the most favourable of the medium corrections above, however, Δ could be as low as 0.49 eV, at the 0.1 % thresholds of Sec. VI: there G is 0.06 % in the reference model and 0.21 % with anisotropic tensors. The estimate therefore places the monoamine oxidase pair outside the range of a measurable field effect unless the active site strengthens the attraction of the pair much more than it desolvates the ions; settling this requires the transfer free energies, or the one-electron flavin potential with substrate bound. Kinetic studies with benzylamine analogues are consistent with this, though they cannot exclude a short-lived pair: no flavin radical is detected during the reductive half-reaction of bovine MAO B,²⁹ and the substituent and isotope effects of human MAO A indicate rate-limiting $\alpha\text{-C-H}$ cleavage by proton abstraction.³⁰ A direct hydride transfer has also been proposed for MAO.²⁸ These estimates concern substrates of the benzylamine type, oxidized at nitrogen. Other substrates, including the catechol and indole neurotransmitters, are not analysed here and would require their own potentials.

The electronic coupling between flavin and substrate does not enter eq 5, but it indicates the regime. Fragment-orbital couplings³¹ for methylamine at van der Waals contact (69 non-clashing geometries of 112; Supporting Information) are 53 meV (median) to 106 meV at 3.5 Å, large enough that back electron transfer at contact is likely to be adiabatic rather than non-adiabatic.³² The reaction terms of eq 2 are derived for weak, non-adiabatic coupling and were tested against exact results for diabatic couplings of up to 3 meV.¹¹ Eq 5 holds for any generator of that form with unital, positivity-preserving relaxation, including the reactive dephasing that appears at

higher order; whether a strongly coupled contact pair is described by such a generator at all is not addressed here, and F is least reliable for it.

VIII. A LIGHT-DRIVEN PAIR IN THE SAME MODEL

The flavin–tryptophan pair of cryptochrome is formed by light,^{3,4} so eq 5 does not apply to it. As a check that the same equations produce a field response when the rates are not constrained, we evaluated it with the same scheme and yield observable, averaged over field orientations rather than maximized, $k_S = k_P = 1 \mu\text{s}^{-1}$, the pair 1.90 nm apart ($D = -11.4$ MHz), flavin N5 and N10 as above, the two largest β -proton couplings (60 and 56 MHz) and N1 of the tryptophan radical cation from the same density-functional method, and the exchange coupling on 21 values spanning the literature range of its prefactor with both signs.⁹ The orientation-averaged response ranges from below $10^{-2}\%$ (it changes sign within the range) to 0.63%. This is a numerical control, not a like-for-like comparison: the hyperfine couplings, dipolar coupling and rates differ from the enzyme model as well as the way the pair is formed.

IX. DISCUSSION

The bound of eq 5 applies to any radical pair that is formed thermally from a closed-shell state, returns to it only through the singlet, and carries the catalytic flux; a pair on a dead-end branch is limited instead by the population bound of the same section. Both require the reaction terms of eq 2 and trace-preserving, unital, positivity-preserving relaxation, but no electronic coupling, reorganization energy or particular onward chemistry. The numbers that follow depend on more: F and G are the results of a finite search in a specific spin model, richer nuclear models were searched in full only with anisotropic flavin tensors, and the monoamine oxidase estimate depends on a dialkylamine potential standing in for the substrates, on how substrate binding shifts the one-electron flavin couple, and on transfer free energies into the active site that are not known.

Two routes around the bound should be named. Three-radical and scavenging mechanisms³³ restore sensitivity to strongly coupled pairs. Whether eq 5 limits them depends on where the scavenging step enters the reaction scheme, and their field effects would have to be computed in their own model. And an active site that brings Δ close to zero is not excluded: competence then allows slow decay and field effects of a percent or more. The question for a particular enzyme is where its radical pair lies in Δ , which is measurable.

X. CONCLUSIONS

For a radical pair that an enzyme forms thermally and through which it turns over, detailed balance bounds the singlet back-transfer rate from below by $4k_{\text{cat}}e^{\Delta/k_B T}$ and the spin-

independent onward rate by $k_{\text{cat}}e^{\Delta/k_B T}$, for any spin Hamiltonian and any trace-preserving, unital, positivity-preserving relaxation. In a flavin–aminium spin model with computed hyperfine and dipolar couplings, the largest field effect found among the searched rates that satisfy these bounds for $k_{\text{cat}} = 1 \text{ s}^{-1}$ falls below 0.1% at $\Delta = 0.47 \text{ eV}$ (0.50 eV with anisotropic hyperfine tensors, 0.47 eV with electron spin relaxation) and stays below up to the largest driving force computed. The measured flavin potentials of the monoamine oxidases, with a dialkylamine proxy for the potential of benzylamine-type substrates, place an amine-to-flavin pair at 0.86–1.44 eV in water, above that range; continuum estimates for the active site leave it there unless the site lowers Δ to about 0.49 eV. The transfer free energies into the active site, or the one-electron flavin potential with substrate bound, would settle the question.

SUPPLEMENTARY MATERIAL

See the supplementary material for the 300-system test of eq 4, the full search results, the electron-spin-relaxation calculations in the four-nucleus model, the hyperfine and dipolar tensors, the coupling scan, the density-functional driving forces, the model-sensitivity calculations, the cryptochrome scan, and software and environment.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Hikaru Wakaura: Conceptualization (lead); Methodology (lead); Software (lead); Validation (lead); Formal analysis (lead); Investigation (lead); Data curation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Taiki Tanimae:** Supervision (lead).

DATA AVAILABILITY

The data that support the findings of this study and the code that generates them are openly available in the qbscreen package at https://github.com/deeptell-inc/brain_protein_screening; every number in the text is generated from the stored results by qbscreen/make_numbers_v6.py.

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